

for a better selection of I/O offerings at this price range.

DISPLAY

The Acer Predator Helios Neo 16 comes with two display variants, 1920x1200 and 2560x1600 resolutions. As you can see, these are 16:10 aspect ratio displays, so you have a little more screen to work with. The variant we received came with a WUXGA or 1920x1200 display. Despite being the cheaper of the two display options, you still get around 400 nits of peak brightness which is great; we tested an average of around 382 nits. Additionally, you get close to 100 percent sRGB coverage, we tested 99.6 percent, and 83.2 percent DCI-P3 coverage, which is great. That makes the Acer Predator Helios Neo 16 a pretty solid laptop for your content creation needs as well.

THERMALS

The Acer Predator Neo 16 performs exceptionally in the thermals department, especially if you consider this is a gaming laptop. Not only do the temps remain low, even the noise is lower than average as well. The highest temperature we recorded was the CPU, which peaked at around 70 degrees celsius during the stress test. Both the CPU and GPU in the Acer Predator Neo 16 use Acer's AeroBlade 3D fans for cooling, and they both share six heat pipes. While the bottom of the laptop can get pretty warm, surface temps stay around the 25 degree mark for the most part, such as around the palm and touchpad, except for the WASD cluster which hits 32 degrees Celsius and the centre of the laptop which goes to around 36 degrees Celsius.

PERFORMANCE

With that we come to the performance. Like we mentioned in the intro, Acer hasn't held back with the hardware in the Neo 16, so we've got the latest 13th Gen Intel Core i7-13700HX paired with the new

NVIDIA RTX 4050. While the RTX 4050 is the lowest end variant in the new RTX 40 series lineup, it's certainly no slouch, and is comparable to the RTX 3060 Laptop GPU when it comes to gaming performance.

Starting things off with the synthetic GPU benchmarks ala 3DMark, we're comparing the 4050 with one of our best performing 3060 laptops from last year, the Lenovo IdeaPad Gaming 3 16IAH7 which also features a 12th gen Core i7.

The 4050 beats the 3060 in just about every 3DMark benchmark save a few, where the scores are very close.

Coming to productivity, for which we use the PCMark 10 benchmark, we're once again comparing the same two laptops, this time pitting the 13th Gen Core i7-13700HX in the Predator Helios Neo 16 with the 12th Gen Core i7-12700H in the Lenovo IdeaPad Gaming 3.

No surprises here, the newer processors wins out. Coming to the Cinebench R23 benchmark, we found that the single-thread performance in both processors is very similar, however the Core i7-13700HX inches ahead of the Core i7-12700H in multi-thread performance.

The trend continues in our encryption and decryption benchmarks as well, for which we use WinRAR and 7-Zip, where the Acer Predator Helios Neo 16 is leagues ahead.

RAM

The Neo 16 comes with 16GB of DDR5 memory clocked at 4800 MT/s with support for up to 32GB. We tested RAM speeds using the AIDA64 memory benchmark, and it's certainly among the more speedier RAMs we've tested this year.

STORAGE

This continues with the NVMe PCIe 4.0 SSD storage on-board as well, which is among the fastest in the price segment. We saw read speeds close to 7000 MB/s and write speeds around the 5000 MB/s mark during the CrystalDiskMark NVMe SSD benchmark.

GAMING PERFORMANCE

Finally, we come to the real-world gaming performance. We use a mix of both GPU and CPU intensive games to gauge how well the Acer Predator Helio Neo 16 performs in real-world gaming. Like we mentioned at the start of this section, the RTX 4050 certainly holds its own. We're once again comparing it with the RTX 3060 in the Lenovo IdeaPad Gaming 3.

The RTX 4050 outperforms the RTX 3060 in every single gaming benchmark. Clearly, the RTX 4050 is the new best entry level gaming laptop GPU. Of course, this specific combination in the Acer Predator Helios Neo 16 is fantastic and should have no trouble handling any modern title you throw at it, at least for the coming year.

VERDICT

The Acer Predator Helios Neo 16 is a very enticing gaming laptop. It offers a high-end gaming laptop experience at a very reasonable budget. You may miss out on a few features here and there, but honestly, who needs per-key RGB lighting or an OLED display when you're getting top-of-the-line hardware onboard. The combination of the 13th Gen Intel Core i7-13700HX and the NVIDIA RTX 4050 is enough to handle most modern games with ease at high settings.

Both the processor and GPU scored fantastically in our benchmarks, and this is paired with a bright 165 Hz WUXGA IPS display, great overall build quality, a decent keyboard, and surprisingly decent battery life as well, for a gaming laptop, that is. At Rs.1,24,990, we certainly wouldn't call this a cheap or a budget gaming laptop, but we can very confidently say that you're getting the best bang for your buck.

—Manish Rajesh

SPECIFICATIONS

Processor: Intel Core i7-13700HX | RAM: 16GB DDR5 | GPU: NVIDIA GeForce RTX 4050 | Storage: 1TB SSD | Display: 16 Inch IPS WUXGA | Refresh Rate: 165 Hz

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